

Cloudhaven Skyport: Balance & Economy

Design workbook by Umut Eren Demirci. Blue cells are inputs; everything else is calculated or measured.

Sheet	What it contains	Live formulas?
Milestones	When three kinds of player reach each moment of the first island, against the pacing targets, with an on-target check.	Target checks
Purchases	Every build pad: its cost, when each player buys it, the wait since the previous purchase, and what it pays back.	Gaps, flags
Income	Coins earned per minute over the island for each player, with a chart.	No
Flow Model	The airport as a pipeline: passengers per minute from the timetable against each counter's capacity. Change the timetable or the staff and see where the bottleneck moves.	Yes
Goals	The goal list, rewards, and when each player meets each goal.	Share of rewards
Campaign	The three islands side by side: money multiplier, twist, and when each player reaches each moment.	No
Monetisation	Offline earnings (the 2-hour cap, the rewarded double) and a day-30 revenue per player from retention, rewarded views and purchases: the numbers behind the CPI go line.	Yes
Tuning	Every number in Tuning.cs, as the build uses it.	No

Source of truth: the game's code (Assets/_Project/Scripts/Core). Measured data comes from the economy simulator (Tools/sim), which compiles those same rules and plays each island 40 times with each of three bots: a new player (slow to decide, explores, pauses, often misses the best move), a casual player and an expert. Times are medians. The tuning story is in Docs/BALANCE.md.

Milestones

Seconds from the start of the island (medians of 40 runs). The target band is for a new player.

Moment	Target from (s)	Target to (s)	New player (s)	Casual (s)	Expert (s)	New player	On target?
First coin	0	10	7	5	5	0:07	on target
First pad (Bigger bag)	0	30	23	18	14	0:23	on target
First staff (customs officer)	120	180	121	103	87	2:01	on target
The Terminal opens	180	420	236	221	225	3:56	on target
The Promenade opens	900	1800	1759	1521	1655	29:19	on target
Every pad bought	1800	3300	3185	2776	2836	53:05	on target
Island complete	2400	3600	3185	2776	2836	53:05	on target

The island is paced for a first-time player to finish in 40 to 60 minutes. Casual and expert players finish sooner mainly because they serve counters by hand (passengers they serve tip 50%) and waste less time; the airship timetable sets the upper limit on income for everyone, which is typical of the genre.

Purchases

Every build pad in the order a new player buys it. Times in seconds (medians); payback from the simulator's income before and after.

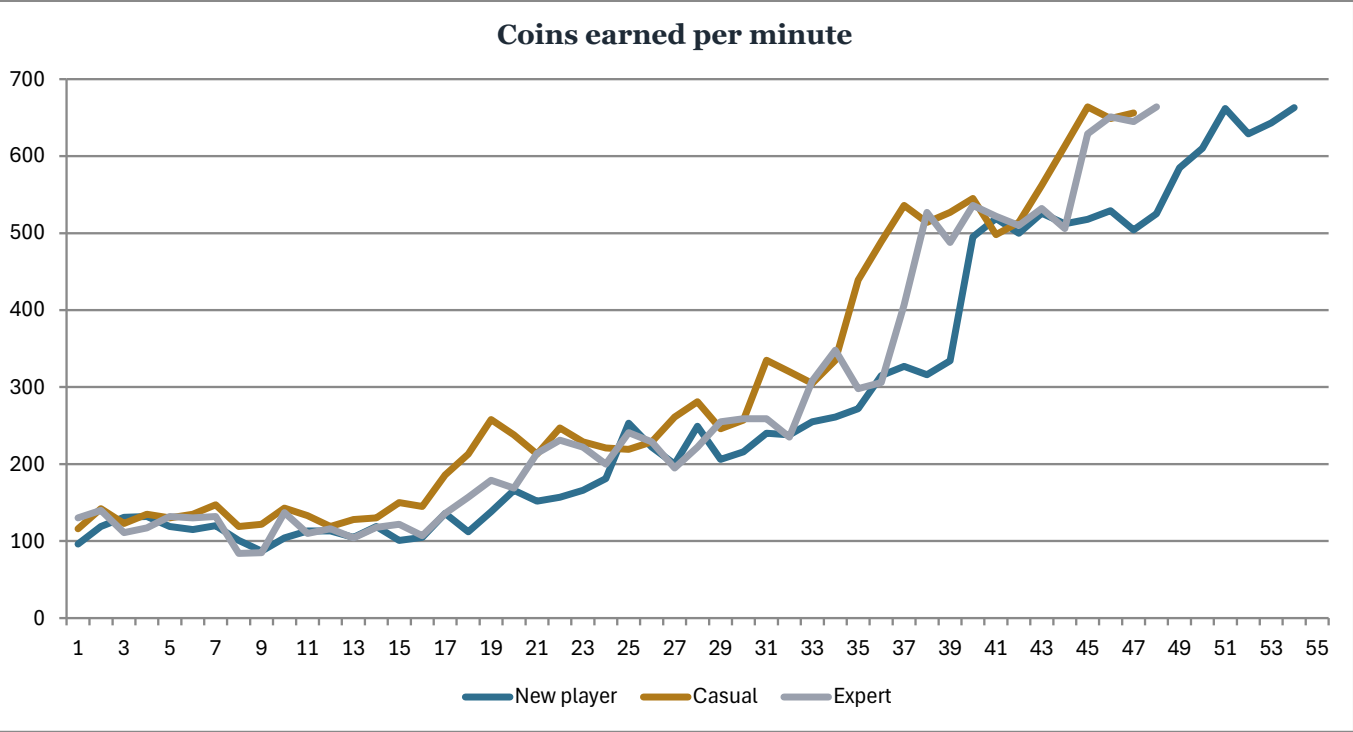
Pad	Cost	New player (s)	Casual (s)	Expert (s)	Wait since previous (s)	Wait over 5 min?	Income before (/min)	Income after (/min)	Payback (min)	Kind
Bigger bag	10	23	18	14	23	no		125	no direct change	Carrying
Customs officer	280	121	103	87	98	no	97	125	10.0	Staff
Open the Terminal	300	236	221	225	115	no	117	116	no direct change	Zone
Security	150	286	266	271	50	no	126	109	no direct change	New counter
Bag drop	220	404	359	370	118	no	122	100	no direct change	New counter
Ticket clerk	300	587	503	548	183	no	102	112	30.0	Staff
Faster customs	250	718	613	669	131	no	108	109	250.0	Capacity
More flights	450	955	823	908	237	no	109	135	17.3	Demand
Guard	400	1102	947	1047	147	no	116	159	9.3	Staff
Porter's cart	300	1227	1040	1148	125	no	138	167	10.3	Carrying
Porter	550	1432	1186	1312	205	no	161	230	8.0	Staff
Faster bag drop	450	1541	1307	1440	109	no	201	215	32.1	Capacity
Open the Promenade	800	1759	1521	1655	218	no	216	247	25.8	Zone
Cafe	500	1761	1523	1657	2	no	216	247	16.1	New counter
Faster security	600	1870	1614	1754	109	no	223	257	17.6	Capacity
Souvenir stall	750	2050	1783	1930	180	no	248	314	11.4	New counter
Bigger airships	1300	2321	2033	2175	271	no	313	514	6.5	Demand
Faster tickets	700	2420	2138	2274	99	no	375	518	4.9	Capacity
Second customs desk	900	2532	2223	2358	112	no	490	518	32.1	Demand
Second security lane	1100	2688	2351	2476	156	no	517	518	1100.0	Capacity
Even more flights	1800	2892	2542	2634	204	no	516	633	15.4	Demand
Grand Liner dock	1800	3185	2776	2836	293	no	639		no direct change	Demand
Total	13,910									

Payback reads the airport as a pipeline. Demand upgrades (more flights, bigger airships) and staff (who free the harbourmaster to earn tips elsewhere) raise income directly and pay back in minutes. Capacity upgrades (faster counters, the second security lane) and new counters mostly keep queues short as demand grows, so they show little change on their own: their value is fewer grumpy passengers, who pay half and skip the shops. Zones open further purchases rather than income.

Income per minute

Coins earned in each minute of the island (medians). A player's line ends when they finish.

Minute	New player	Casual	Expert
1	96	116	130
2	119	142	140
3	131	123	111
4	132	135	117
5	119	130	132
6	115	135	130
7	120	147	132
8	101	119	84
9	87	122	85
10	104	143	137
11	113	133	110
12	113	119	116
13	105	128	104
14	119	130	118
15	101	150	122
16	105	145	107
17	136	186	136
18	112	213	157
19	138	258	179
20	166	238	169
21	152	213	214
22	157	247	231
23	166	229	222
24	181	221	200
25	253	219	241
26	222	229	229
27	200	261	195
28	249	281	222
29	206	246	255
30	216	257	259
31	240	335	259
32	238	320	235
33	255	305	308
34	261	335	348



35	272	439	298
36	315	489	306
37	327	536	407
38	316	514	527
39	334	527	488
40	495	545	536
41	519	498	522
42	500	514	510
43	526	562	532
44	512	613	506
45	518	664	629
46	529	649	651
47	504	656	645
48	525		664
49	585		
50	610		
51	662		
52	629		
53	643		
54	663		
55			

The airport as a pipeline

Change a blue cell: passengers per minute (demand) against each counter's capacity. Utilisation over 100% means queues grow and passengers turn grumpy.

The timetable

Seconds between airships	28
Passengers per airship (each way)	10
A Grand Liner every ... ships	3
Liner size (x)	2
Arrivals per minute	28.6
Departures per minute	28.6

Counters

Counter	Flow	Seconds to serve	Fare	Staffed? (1/0)	Speed x	Harbourmaster here? (1/0)	Capacity (/min)	Demand (/min)	Utilisation	Income (/min)
Customs	Arrivals	1.4	5	1	2	0	38.6	28.6	74%	143
Tickets	Departures	1.1	4	1	1.5	0	36.8	28.6	78%	114
Bag drop	Departures	1.2	3	1	1.5	0	33.8	28.6	85%	86
Security	Departures	1.5	3	1	2	0	36.0	28.6	79%	86
Cafe	Shop	2.2	10	1	1	0	27.3	12.9	47%	129
Souvenirs	Shop	2.6	15	1	1	0	23.1	12.9	56%	193
Total										750

Staff pace (x the harbourmaster)	0.45
Shopkeeper pace	1
Chance a passenger visits each shop	0.45
Service tip when the harbourmaster serves (x)	1.5

The late island is shown: every counter staffed, every speed upgrade bought, airships every 28 s with 10 passengers each way and a Grand Liner every third ship (the opening timetable is 40 s, 6 passengers, a liner every fifth). Before the second customs desk was added, customs sat at 99% here: this sheet found that late-game bottleneck. The lesson of the tuning: with staff at 45% of the harbourmaster's pace, a counter handles about 20 passengers a minute; demand passes that once the flight upgrades land, so every flight upgrade is paired with capacity (faster counters, a second security lane), or passengers turn grumpy and the income falls, which the first tuning pass showed. Full-ship bonuses are not included.

Goals

Three goals are shown at a time, in order; each reward goes into the harbourmaster's bag. Times in seconds (medians).

Goal	Reward	New player (s)	Casual (s)	Expert (s)	Share of all rewards
Sell 10 tickets	20	88	78	70	1%
Hire a customs officer	40	121	103	87	2%
Open the Terminal	60	236	221	225	3%
Send off a full airship	80	102	84	70	4%
Screen 60 passengers at security	120	993	833	923	5%
Open the Promenade	200	1759	1521	1655	9%
Serve 50 coffees	250	2376	2097	2254	11%
Earn 10000 coins	400	2701	2429	2571	18%
Send off 15 full airships	500	1759	1521	1655	22%
Build the Grand Liner dock	600	3185	2776	2836	26%

The campaign

Three islands with the same 22 pads and 10 goals; money scales, and a twist changes what matters. Times are medians (m:ss).

Island	Money (x)	Twist	Moment	Target	New player	Casual	Expert
Cloudhaven		1 None: learn the ropes	First coin	0:00-0:10	00:07	00:05	00:05
Cloudhaven		1 None: learn the ropes	First pad (Bigger bag)	0:00-0:30	00:23	00:18	00:14
Cloudhaven		1 None: learn the ropes	First staff (customs officer)	2:00-3:00	02:01	01:43	01:27
Cloudhaven		1 None: learn the ropes	The Terminal opens	3:00-7:00	03:56	03:41	03:45
Cloudhaven		1 None: learn the ropes	The Promenade opens	15:00-30:00	29:19:00	25:21:00	27:35:00
Cloudhaven		1 None: learn the ropes	Every pad bought	30:00-55:00	53:05:00	46:16:00	47:16:00
Cloudhaven		1 None: learn the ropes	Island complete	40:00-60:00	53:05:00	46:16:00	47:16:00
Galewind Isle		2 Storms: every 4th ship 12 s late, x1.5 passengers; mirrored	First coin	0:00-0:10	00:07	00:05	00:05

Galewind Isle	2 Storms: every 4th ship 12 s late, x1.5 passengers; mirrored	First pad (Bigger bag)	0:00-0:30	00:26	00:18	00:12
Galewind Isle	2 Storms: every 4th ship 12 s late, x1.5 passengers; mirrored	First staff (customs officer)	2:00-3:00	02:37	01:53	01:32
Galewind Isle	2 Storms: every 4th ship 12 s late, x1.5 passengers; mirrored	The Terminal opens	3:00-7:00	05:13	03:51	03:37
Galewind Isle	2 Storms: every 4th ship 12 s late, x1.5 passengers; mirrored	The Promenade opens	15:00-30:00	34:38:00	28:06:00	28:20:00
Galewind Isle	2 Storms: every 4th ship 12 s late, x1.5 passengers; mirrored	Every pad bought	30:00-55:00	58:43:00	50:01:00	48:37:00

Galewind Isle	2 Storms: every 4th ship 12 s late, x1.5 passengers; mirrored	Island complete	40:00-60:00	58:43:00	50:01:00	48:37:00
Lantern Reach	4 VIPs: 15% pay 3x, with half the patience	First coin	0:00-0:10	00:07	00:05	00:05
Lantern Reach	4 VIPs: 15% pay 3x, with half the patience	First pad (Bigger bag)	0:00-0:30	00:24	00:17	00:12
Lantern Reach	4 VIPs: 15% pay 3x, with half the patience	First staff (customs officer)	2:00-3:00	02:26	01:45	01:13
Lantern Reach	4 VIPs: 15% pay 3x, with half the patience	The Terminal opens	3:00-7:00	04:39	03:31	02:47
Lantern Reach	4 VIPs: 15% pay 3x, with half the patience	The Promenade opens	15:00-30:00	29:35:00	26:20:00	25:00:00

Lantern Reach	4 VIPs: 15% pay 3x, with half the patience	Every pad bought	30:00-55:00	51:01:00	44:50:00	42:08:00
Lantern Reach	4 VIPs: 15% pay 3x, with half the patience	Island complete	40:00-60:00	51:01:00	44:50:00	42:08:00

Prices, fares, goal rewards, carry capacity, coin pick-up speed and pad pay-in speed all scale with the island's money multiplier. Scaling prices alone made Galewind Isle 75 minutes long and Lantern Reach unfinishable in 90. Galewind's Promenade opens just past the 30-minute target: storms leave fewer airships full. Lantern Reach is the fastest: VIP fares outweigh their shorter patience.

Offline earnings and revenue per player

Change a blue cell. The offline rules are the game's (Idle.cs); the revenue model is a planning assumption for the CPI test.

Offline earnings

Island's played income (coins/min)	630
Offline efficiency (share of played rate)	30%
Cap (hours)	2
Hours away	8
Coins waiting on the dock	22,680
With the rewarded double	45,360
Minutes of play the doubled coins are worth	72

Revenue per player by day 30

D1 retention	40%
D7 retention	12%
D30 retention	4%
Active days in the first 30 (day 0 counts)	4.5
Rewarded views per active day	2.5
Rewarded eCPM (US, \$)	\$12.00
Interstitials per active day (0 = as designed)	0
Interstitial eCPM (\$)	\$8.00
Payers (share of installs)	1.0%
Average purchase (\$, after store fee)	\$2.10
Ad revenue per active day (\$)	\$0.030
Revenue per player by day 30 (\$)	\$0.157
CPI that pays back by day 30 (\$)	\$0.157

As designed the game shows only rewarded ads, so a player earns about \$0.16 by day 30; the CPI test's go line of \$0.35 needs payback after day 30 and a first monetisation test in soft launch. Set interstitials to 3 a day (shown only on leaving an island) to see that test's alternative: about \$0.27. Retention between the given days is interpolated linearly, which slightly overstates the tail.

Tuning

Every field of *Tuning.cs*, as the build uses it. Station arrays: Customs, Tickets, Bag drop, Security, Cafe, Souvenirs.

Field	Value
PlayerSpeed	5.5
ServeRadius	1.3
CollectRadius	1.2
CollectRate	30
CarryCapacity	30
CarryUpgradeScale	1
PadRadius	1.1
PadDrainTime	1.2
PadMinRate	20
PassengerSpeed	2.4
QueueSpacing	0.75
Patience	40
GrumpyFare	0.5
ShopVisitChance	0.45
ServiceTime	1.4 / 1.1 / 1.2 / 1.5 / 2.2 / 2.6
Fare	5 / 4 / 3 / 3 / 10 / 15
StaffPace	0.45
ServiceTip	1.5
ShopPace	1
SpeedMultiplier	1 / 1 / 1 / 1 / 1 / 1
ShipInterval	40
FirstShipDelay	3
ShipApproachTime	3
DockTime	16
ShipPassengers	6
Seats	6
DisembarkGap	0.6
BoardGap	0.5
FullShipBonusPerSeat	3
LinerEvery	5
LinerSize	2
StormEvery	0
StormDelay	12
StormSize	1.5
VipChance	0
VipFare	3
VipPatience	0.5
DeparturesPerShip	6
FirstDepartureDelay	1
MaxDeparturesWaiting	18